

MS4 Finance

1)

(a) $£1230 \times 0.09 = £111$

(b)

$$P = \frac{£110}{1.08} + \frac{£110}{1.08^2} + \dots + \frac{£130}{1.08^7} = £1294$$

2) $PV = \frac{C}{(1+r)^t}$

Years 1-5

$$\frac{0.65}{1.08} + \frac{0.73}{1.08^2} + \frac{0.82}{1.08^3} + \frac{0.91}{1.08^4} + \frac{1.02}{1.08^5} \approx 3.24$$

Year 5 onwards

$$\frac{1.02 \times 1.02}{0.8 - 0.02} = 17.387$$

Share value

$$3.24 + \frac{17.387}{1.08^5} = 15.0735 = £15.07$$

In this question I have ~~also~~ summed the discounted cash flows from years 1-5, then added this to the present value of the constant growth dividend from year 5 onwards.

3)

(a) Because the dividend is constant, and we assume the company will exist for the foreseeable future, we can use the ~~annuity~~ ^{perpetuity} calculation to value the share

$$PV = \frac{C_1}{r} \quad \$150 = \frac{\$12}{r} \quad r = 0.08 \quad r = 8\%$$

N.B. this assumes the market (share price) accurately represents the PV of the share, which implicitly assumes well-informed investors.

- (b) Here we would use the Gordon growth model to calculate the value of a share with constant dividend growth.

$$P_0 = \frac{D_{n+1}}{r-g} \quad \$150 = \frac{\$12 \times 1.05}{r-0.05}$$

$$\$150r - 7.5 = 12.6 \quad 150r = 20.1 \quad r = 0.134 \quad r = 13.4\%$$

Intuitively this makes sense; we would expect a growing-dividend share to be more valuable than a constant dividend share, ceteris paribus. If they are of the same value, a higher discount rate (cost of equity) for the constant growth dividend could explain this, and $13.4\% > 8\%$

4.

(a) $\frac{€8}{0.08} = €100$

(b) $\frac{€8 \times 1.02}{0.08 - 0.02} = €136$

(c) $\frac{€8 \times 1.15}{0.08 - 0.15} = -€131$

→ Gordon constant growth model doesn't work when $g > r$, but not sure why?

5) $r = 0.1 \quad D_0 = 2.1 \quad D_1 = 2.415 \quad g = 0.04$

Years 1-3 PV

$$\frac{2.415}{1.1} + \frac{2.77725}{1.1^2} + \frac{3.19384}{1.1^3} = €6.89$$

Year 3 onwards

$$\frac{3.19384 \times 1.04}{0.1 - 0.04} = €55.35989$$

Overall share PV

$$£6.89 + £55.36 = £62.25$$

$$\frac{£6.89 + £55.36}{1.1^3} = £48.48$$

6)

$$NPV = -150m + \frac{20}{0.1}$$

NPV = 50 $50 > 0$, So invest at this discount rate if there are no mutually exclusive projects with a higher NPV

$$\text{Need } \frac{20}{r} \geq 150$$

$$20/0.15 = 133.33$$

$$20/0.125 = 160$$

$$20/0.13 = 153.35$$

$$20/0.133 = 150.37$$

$$20/0.134 = 149.25$$

Discount rate of 13.3% gives a positive NPV and discount of 13.4% gives a negative NPV, So IRR $\approx$ 13.3%. This means the maximum allowable deviation in the cost of capital is 3.3 percentage points.

7) Year 0 outflows

- 520,000 on cars

- 200,000 on signage

- 720,000 total

Years 1-4 netflows

+ 200,000 car sales (30k-26k x 50)

- 30,000 employee's salary

- 10,000 employee bonus

160,000 net flow per year

96,000 per year after tax

Year 5 net flows

96,000 after-tax car sales profit

50,000 tax free equipment sale

312,000 after-tax working capital recovery

458,000

→ Unsure whether working capital sales should be taxed here?

$$\text{Thus NPV} = \frac{-720,000}{1.12} + \frac{96,000}{1.12} + \frac{458,000}{1.125}$$

$$= -£163,532.96 < 0, \text{ so don't invest}$$

→ Unsure at what point to apply tax rate

→ Unsure whether working capital sales should be assumed as same price as purchase

$$8) \text{ WACC} = \frac{E}{V} r_e + \frac{D}{V} r_d (1 - T_c)$$

D and E are market values of firm's debt and equity, V is total market value of the firm. r_e and r_d are costs of equity and debt, $(1 - T_c)$ is the 'tax shield'.

$$\text{WACC} = \frac{8}{11} \times 0.12 + \frac{3}{11} \times 0.07 (0.6)$$

$$\text{WACC} = \frac{24}{275} + \frac{63}{5500}$$

$$\text{WACC} = 0.09872$$

$$\text{WACC} = 9.9\%$$

This is the rate at which we would discount a project's future cash flows to calculate NPV